

A Level Mathematics 9709 · P1 Pure Mathematics 1

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 9 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

目录 | Contents

1. 二次函数 | Quadratics
2. 函数与图像 | Functions and graphs
3. 坐标几何 | Coordinate geometry
4. 弧度与扇形 | Circular measure
5. 三角函数与方程 | Trigonometry
6. 数列、二项式 | Series and binomial
7. 微分基础 | Differentiation
8. 积分基础 | Integration
9. P1 综合建模 | P1 problem solving

第 1 课 | 二次函数 | Quadratics

Lesson 1 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

完成平方揭示顶点，判别式揭示实根数，直线与抛物线联立仍归结为二次方程。

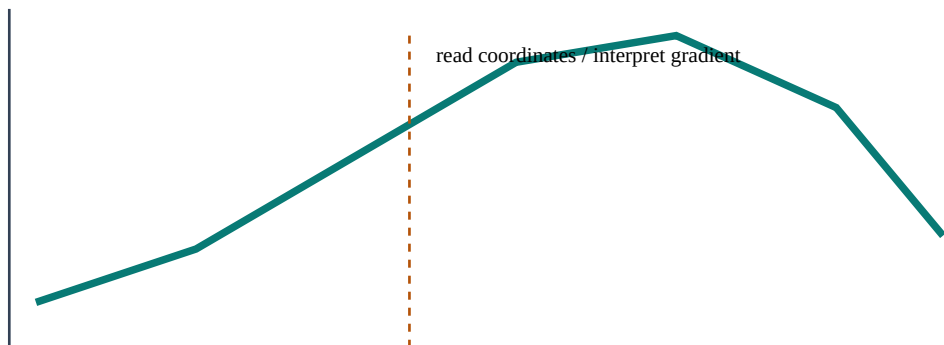
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

把 $ax+bx+c$ 写成 $a(xh)+k$ ；切线条件是联立后判别式为0。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

将 $2x^2+7x+7$ 写成完成平方形式并求最小值。

$2(x+3)^2+11$ ，最小值11。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | 二次函数 | Quadratics

例题二：Cambridge 风格应用 | Cambridge-style application

Find k if $y=2x+k$ is tangent to $y=x$.

$x=2x+k$ gives $x^2-x+k=0$. Tangency: discriminant $4-4k=0$, so $k=1$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

完成平方揭示顶点，判别式揭示实根数，直线与抛物线联立仍归结为二次方程。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。把 $ax+bx+c$ 写成 $a(xh)+k$ ；切线条件是联立后判别式为0。

Detailed English explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：将 $2x^2+7x+7$ 写成完成平方形式并求最小值。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $2(x+3/4)^2+11/2$ ，最小值 $11/2$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find k if $y=2x+k$ is tangent to $y=x^2$.

独立完成后对照： $x^2=2x+k$ gives $x^2-2x-k=0$. Tangency: discriminant $4+4k=0$, so $k=-1$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 函数与图像 | Functions and graphs

Lesson 2 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

定义域决定允许输入，值域决定可能输出；复合与反函数要求正确次序和一一对应。

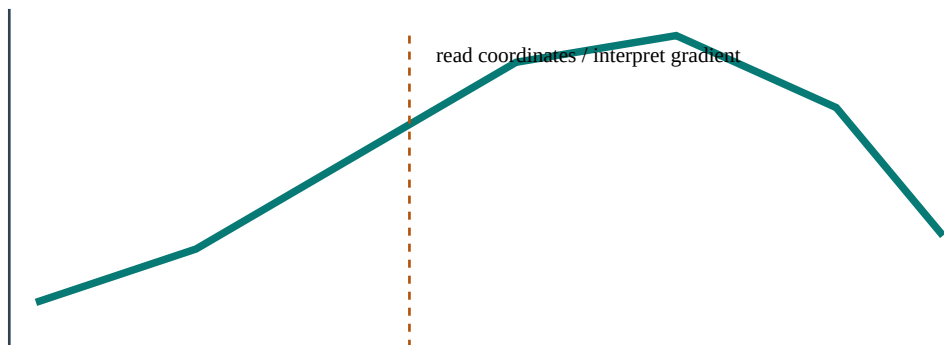
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

做变换时 $y=f(x)+a$ 竖移, $y=f(xa)$ 右移; 伸缩必须观察括号内外。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$f(x)=2x+3$, $g(x)=x+1$, 求 $gf(2)$ 。

$f(2)=7$, 再 $g(7)=8$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 2 课（续） | 函数与图像 | Functions and graphs

例题二：Cambridge 风格应用 | Cambridge-style application

State the transformation from $y=f(x)$ to $y=f(2x)$.

A horizontal stretch with scale factor $1/2$ (compression towards the y -axis).

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

定义域决定允许输入，值域决定可能输出；复合与反函数要求正确次序和一一对应。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。做变换时 $y=f(x)+a$ 竖移， $y=f(xa)$ 右移；伸缩必须观察括号内外。

Detailed English explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $f(x)=2x3$ ， $g(x)=x+1$ ，求 $gf(2)$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $f(2)=1$ ，再 $g(1)=2$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

State the transformation from $y=f(x)$ to $y=f(2x)$.

独立完成后对照： A horizontal stretch with scale factor $1/2$ (compression towards the y -axis).

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 坐标几何 | Coordinate geometry

Lesson 3 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

直线由梯度与一点确定；圆的标准式直接给圆心与半径。

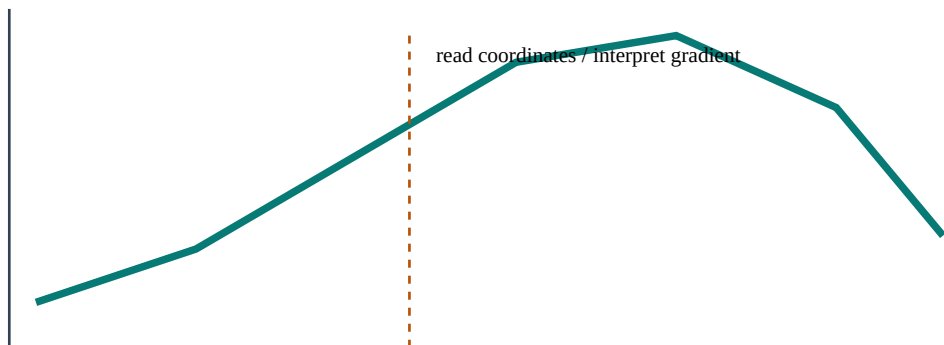
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

垂直梯度乘积1；圆切线垂直于该点半径。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

求圆 $x^2+y^2+6x+4y-12=0$ 的圆心半径。

$(x+3)^2+(y+2)^2=25$, 圆心 $(-3,-2)$, 半径5。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 3 课（续） | 坐标几何 | Coordinate geometry

例题二：Cambridge 风格应用 | Cambridge-style application

Find the tangent gradient at (6,2) to a circle centred at (2,4).

Radius gradient = $(4-2)/(2-6) = 1/2$; tangent gradient = 2.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

直线由梯度与一点确定；圆的标准式直接给圆心与半径。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。垂直梯度乘积1；圆切线垂直于该点半径。

Detailed English explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求圆 $x^2+y^2+6x+4y+12=0$ 的圆心半径。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：(x3)+(y+2)=25，圆心(3,2)，半径5。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find the tangent gradient at (6,2) to a circle centred at (2,4).

独立完成后对照：Radius gradient=(24)/(62)=1/2; tangent gradient=2.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课 | 弧度与扇形 | Circular measure

Lesson 4 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

弧度用弧长与半径的比定义；公式 $s=r\theta$ 、 $A=\frac{1}{2}r^2\theta$ 要求 θ 为弧度。

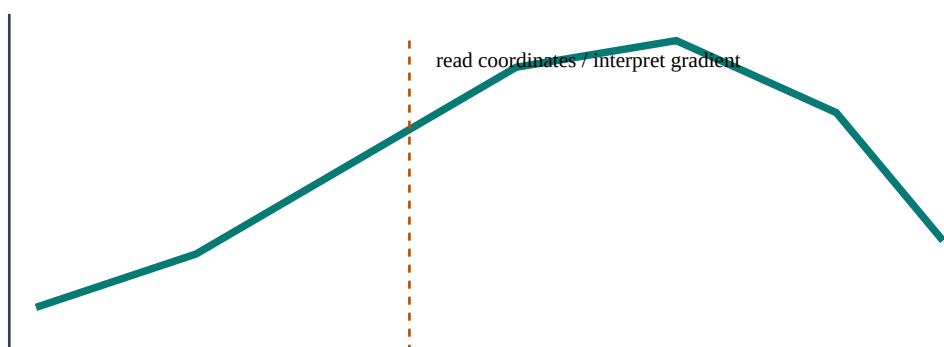
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

角度转弧度乘 $\pi/180$ ；周长题别漏两条半径。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

半径8、角1.2 rad，求弧长与扇形面积。

$s=9.6$ ； $A=0.5 \times 64 \times 1.2=38.4$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续） | 弧度与扇形 | Circular measure

例题二：Cambridge 风格应用 | Cambridge-style application

Convert 135° to radians.

$$135 \pi / 180 = 3 \pi / 4.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

弧度用弧长与半径的比定义；公式 $s=r\theta$ 、 $A=\frac{1}{2}r^2\theta$ 要求 θ 为弧度。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。角度转弧度乘 $\pi/180$ ；周长题别漏两条半径。

Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：半径8、角1.2 rad，求弧长与扇形面积。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $s=9.6$ ； $A=0.5 \times 64 \times 1.2=38.4$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Convert 135° to radians.

独立完成后对照： $135 \pi / 180 = 3 \pi / 4$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 三角函数与方程 | Trigonometry

Lesson 5 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

单位圆解释正负与周期；方程必须在给定区间内找全部解。

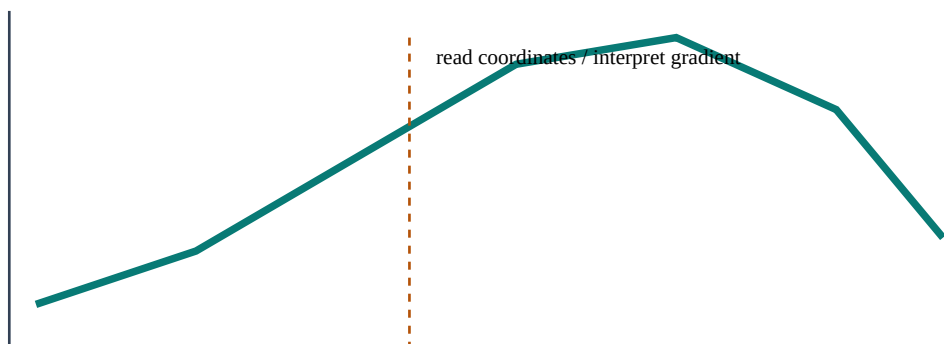
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

先求参考角，再用象限符号确定解；注意degree/radian模式。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

解 $2\sin x = 1$, $0^\circ \leq x \leq 360^\circ$ 。

$\sin x = 0.5$, $x = 30^\circ, 150^\circ$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续） | 三角函数与方程 | Trigonometry

例题二：Cambridge 风格应用 | Cambridge-style application

Solve $\cos(2x)=0$ for $0 \leq x \leq \pi$.

$2x = \pi/2$ or $3\pi/2$, so $x = \pi/4$ or $3\pi/4$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

单位圆解释正负与周期；方程必须在给定区间内找全部解。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先求参考角，再用象限符号确定解；注意degree/radian模式。

Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：解 $2\sin x=1$, $0^\circ \leq x \leq 360^\circ$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $\sin x=0.5$, $x=30^\circ, 150^\circ$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Solve $\cos(2x)=0$ for $0 \leq x \leq \pi$.

独立完成后对照： $2x=\pi/2$ or $3\pi/2$, so $x=\pi/4$ or $3\pi/4$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 数列、二项式 | Series and binomial

Lesson 6 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

等差数列有固定差，等比数列有固定比；二项式系数由组合数产生。

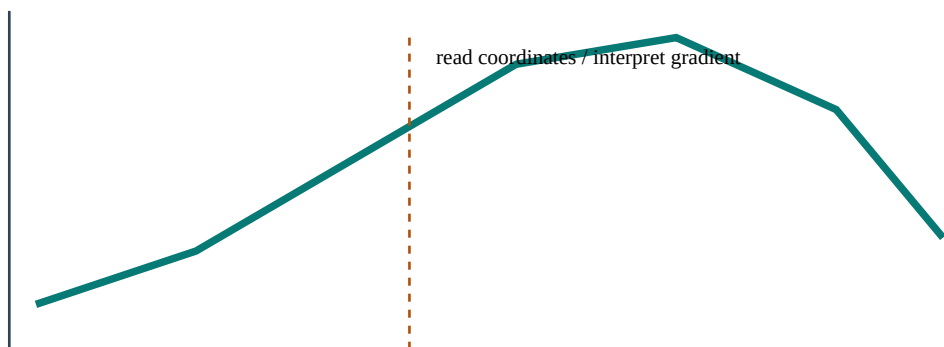
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

AP: $un=a+(n-1)d$; GP: $un=ar^{(n-1)}$; 有限和公式按条件选择。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

等差数列首项5、公差3，求前20项和。

$$S_{20} = \frac{20}{2}[2 \times 5 + 19 \times 3] = 670.$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续） | 数列、二项式 | Series and binomial

例题二：Cambridge 风格应用 | Cambridge-style application

Find coefficient of x in $(1+3x)^5$.

$C(5,2)(3x)=10 \times 9x$, so coefficient 90.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

等差数列有固定差，等比数列有固定比；二项式系数由组合数产生。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。AP: $un=a+(n-1)d$; GP: $un=ar^{(n-1)}$; 有限和公式按条件选择。

Detailed English explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：等差数列首项5、公差3，求前20项和。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $S_{20}=20/2[2 \times 5+19 \times 3]=670$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find coefficient of x in $(1+3x)^5$.

独立完成后对照： $C(5,2)(3x)=10 \times 9x$, so coefficient 90.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | 微分基础 | Differentiation

Lesson 7 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

导数是瞬时变化率和切线梯度；驻点满足 $dy/dx=0$ 。

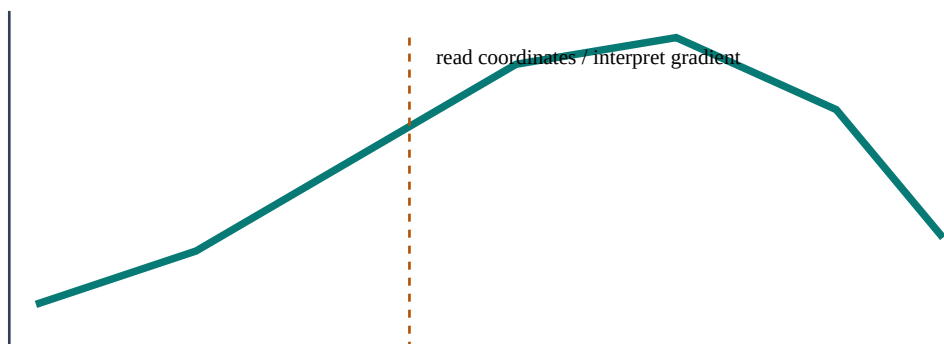
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

幂函数逐项微分；用二阶导或梯度变号分类。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$y=x^6x+9x$ ，求驻点。

$y' = 3 \times 12x + 9 = 3(x+3)$ ， $x=-3$ ；代回得 $(-3,0)$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续） | 微分基础 | Differentiation

例题二：Cambridge 风格应用 | Cambridge-style application

Find the normal to $y=x$ at $x=2$.

Tangent gradient=4, normal gradient=1/4; through (2,4): $y-4=(x-2)/4$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

导数是瞬时变化率和切线梯度；驻点满足 $dy/dx=0$ 。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。幂函数逐项微分；用二阶导或梯度变号分类。

Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $y=x^6+9x$ ，求驻点。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $y'=3x^{12}+9=3(x^1)(x^3)$ ， $x=1,3$ ；代回得 $(1,4),(3,0)$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find the normal to $y=x$ at $x=2$.

独立完成后对照：Tangent gradient=4, normal gradient=1/4; through (2,4): $y-4=(x-2)/4$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 8 课 | 积分基础 | Integration

Lesson 8 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

不定积分是导数的逆运算并带常数C；定积分给有符号面积。

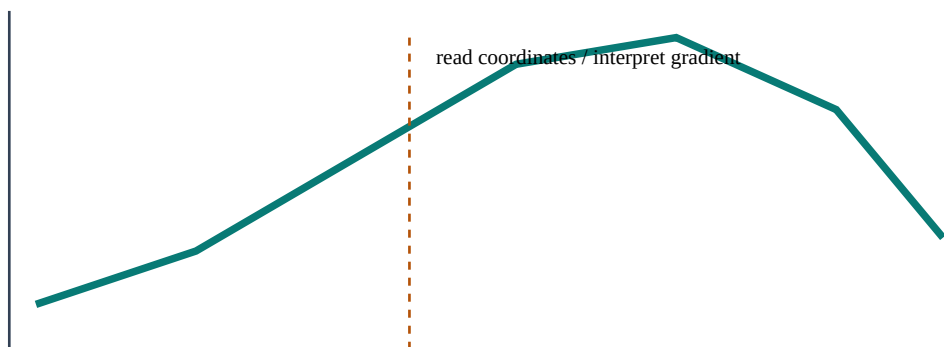
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

x 积分为 $x^{(n+1)/(n+1)}$, $n \neq -1$ ；面积跨 x 轴要分段取正。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

求 $\int (3x^4 + 2) dx$ 。

$x^2 + 2x + C$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 8 课（续） | 积分基础 | Integration

例题二：Cambridge 风格应用 | Cambridge-style application

Find area under $y=2x+1$ from $x=0$ to $x=3$.

$[x+x]=12$ square units.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

不定积分是导数的逆运算并带常数C；定积分给有符号面积。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 x^n 积分为 $\frac{x^{n+1}}{n+1}$ ， $n \neq -1$ ；面积跨x轴要分段取正。

Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求 $\int (3x^4x+2)dx$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $x^2x+2x+C$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find area under $y=2x+1$ from $x=0$ to $x=3$.

独立完成后对照： $[x+x]=12$ square units.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 9 课 | P1 综合建模 | P1 problem solving

Lesson 9 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

综合题常把函数、坐标、三角或微积分串联；每一步都要保留精确值。

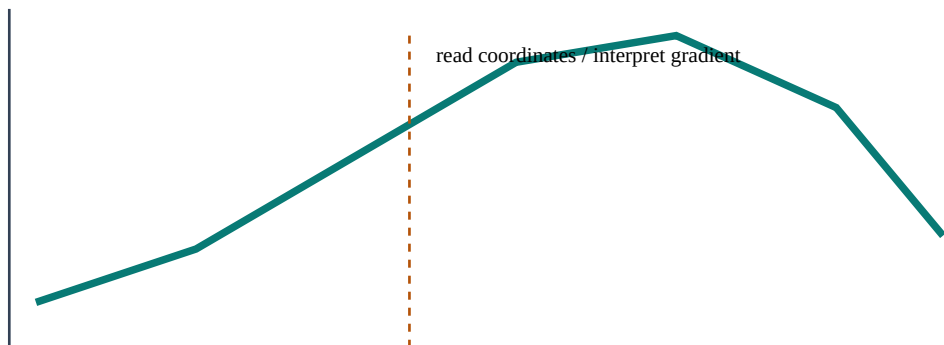
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

先定义变量并画草图，分小问建立结果链；最后检查定义域与几何意义。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

曲线 $y=x^4+7$ 在 $x=1$ 处切线与坐标轴围成三角形，求面积。

点(1,4)，梯度2，切线 $y=2x+6$ ；截距3和6，面积=9。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 9 课（续） | P1 综合建模 | P1 problem solving

例题二：Cambridge 风格应用 | Cambridge-style application

Explain why exact values should be retained.

Premature rounding propagates error into later parts; use exact forms or stored calculator values until the final line.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

综合题常把函数、坐标、三角或微积分串联；每一步都要保留精确值。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先定义变量并画草图，分小问建立结果链；最后检查定义域与几何意义。

Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：曲线 $y=x^4x+7$ 在 $x=1$ 处切线与坐标轴围成三角形，求面积。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：点(1,4)，梯度2，切线 $y=2x+6$ ；截距3和6，面积=9。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Explain why exact values should be retained.

独立完成后对照：Premature rounding propagates error into later parts; use exact forms or stored calculator values until the final line.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?